RISE OMR — Template Design System v5

Architecture for a fully configurable, dashboard-driven OMR engine. RISE 2027 is the first exam config — not a fixed template.

What changed in this pass: Header rebuilt as a branded blue banner with an editable logo slot and school name "RK ENGLISH MEDIUM SCHOOL" · 🟦 blue + 🟧 orange brand palette applied to header, section bars and accents · Phone Number moved into its own aligned line · Paper Set A–D rebuilt as evenly spaced bubbles matching the answer-bubble size · the QR badge box is now removed from the visible wireframe entirely — titles re-centered across the full banner — while machine identification (template/version/sheet/set) still exists in the template schema, generated onto the real PDF at render time rather than hand-drawn in this mockup.

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1. System Flow

Dashboard → Database → Exam Config → Section Config → Layout Engine (auto-reflow) → Template Config (JSON, versioned) → PDF Generator → Print → RICOH IM 2700 Scan → OMR Recognition Engine → Marking Engine (configurable rules) → Results

> Hard rule: the OMR sheet only **records** answers (bubble grid + registration marks + QR). Marking logic lives entirely in the evaluation engine, never in the physical layout.

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2. Marks ≠ Question Count

Section marks, question count, and marks-per-question are three independent numbers stored separately — none derived from another. Validation only checks `Σ(question marks) == allocated_marks`; it never infers question count from marks, and — since the last revision — marks don't print on the sheet at all, so there's nothing to misread.

3. Color System

Two-color brand palette, used deliberately so it never touches the scanner-critical areas:

Element	Color	Why
Header banner background	Blue #123E7C	Brand identity, top of sheet only
Header accents, "OMR ANSWER SHEET" label, section accent strip	Orange #E8720C	Visual hierarchy, warmth against the blue
Section header bar fill	Light blue tint #EAF1FB	Section separation, still >90% light so contrast stays scanner-safe
Answer bubbles, registration marks, question numbers	Pure black on white	These are what the scanner samples — color never applied here, by design

Both brand colors are stored as `brand_colors.primary` / `brand_colors.accent` in the template config, so a different school on the same platform gets its own palette without touching the layout engine.

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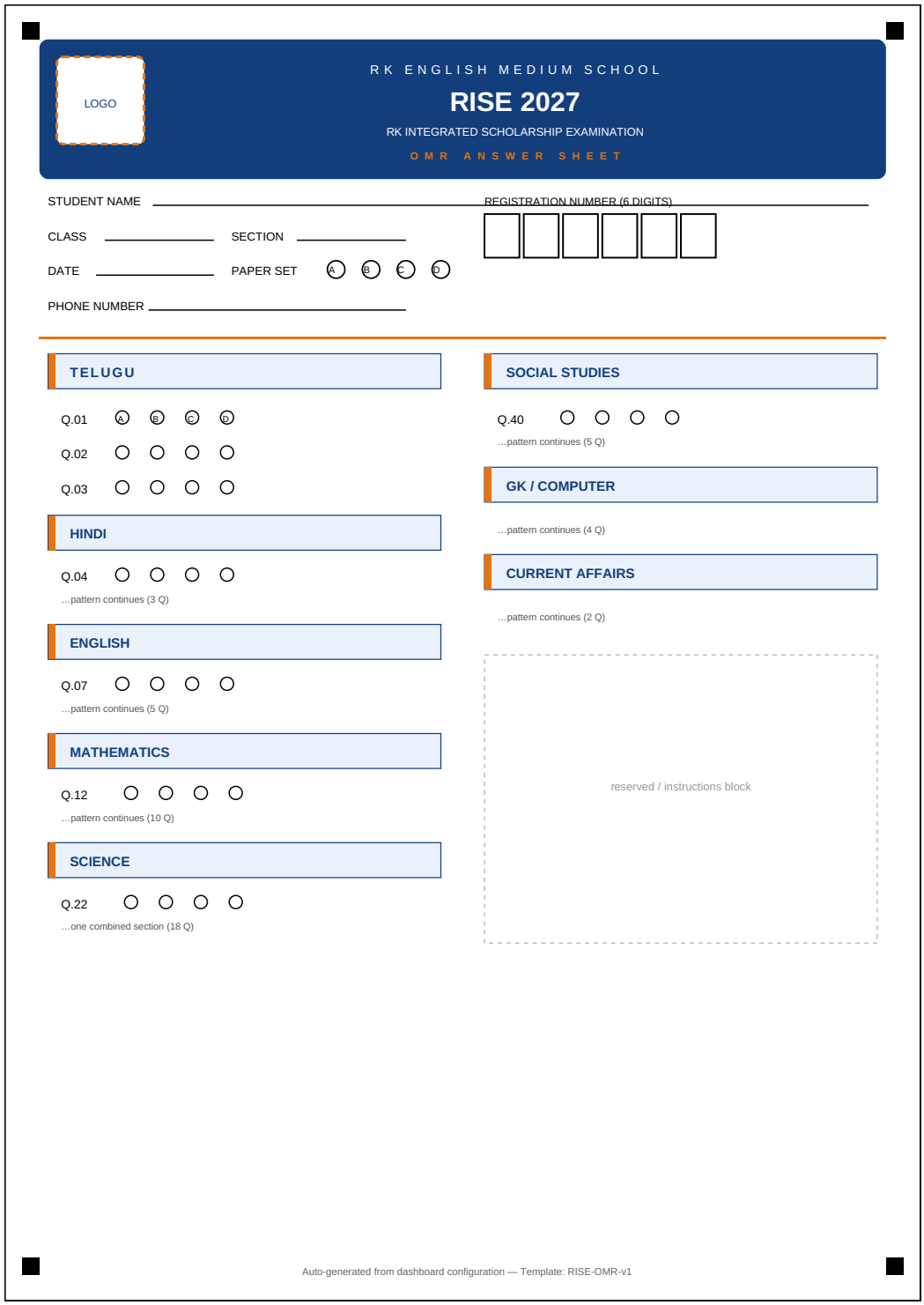
4. Wireframe — A4 Portrait **structural, not final art**

Telugu is drawn in full to show the repeating unit; other sections show one sample row.



5. Section Header Hierarchy

Header bar = light-blue tint fill, thin orange accent strip on the left edge, section name in bold navy — no marks shown anywhere on the sheet. `allocated_marks` still lives in config and drives validation + the marking engine, it just never prints.

6. Student Information Block

Field	Type	Notes
Student Name	handwritten, full width	required
Class	handwritten	required
Section	handwritten	required
Date	handwritten	required
Paper Set	OMR bubble (A/B/C/D), 4mm diameter — same size/spacing as answer bubbles	enable/disable/rename sets from dashboard
Phone Number	handwritten, own line, no longer sharing space with the QR	optional by default, configurable
Registration Number	6 handwritten boxes	required, no digit bubbles

All rows are `StudentField` records — add, edit, remove, reorder from the dashboard. Same shape covers future fields (Parent Name, Candidate ID, Centre Code).

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7. Header / Branding — Fully Editable

Everything in the blue banner is a template field, not hard-coded art:

Field	Editable from dashboard
Logo	Upload PNG/SVG → fills the logo slot at fixed position/size
School name	Free text — "RK ENGLISH MEDIUM SCHOOL" for RISE, anything for the next exam
Exam title / subtitle	Free text — "RISE 2027", "RK INTEGRATED SCHOLARSHIP EXAMINATION", "OMR ANSWER SHEET"
Brand colors	`primary` / `accent` hex — swap blue/orange for any school's colors
Extra header line	Optional subtitle slot (year, batch, centre) — on/off

8. Machine Identification — No Visible QR Box

The wireframe no longer draws any QR box on the sheet. What it was drawing before was a decorative placeholder anyway — a real QR is generated by the PDF renderer at export time from `qr.encodes` (template_id, version, exam_id, sheet_id, paper set), not hand-illustrated in a mockup. That still leaves a decision for the actual PDF: either place a real generated QR somewhere unobtrusive (small, in a margin/corner, outside the header banner) or drop machine-scanned identification altogether and rely on the registration number + manual sheet sorting. The schema keeps `qr` as an optional block either way — `qr.enabled: false` turns it off cleanly without touching layout logic.

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9. Registration Marks — 4 Marks

Top-left, top-right, bottom-left, bottom-right — pure black 4×4mm squares, positioned outside the branded banner in the plain white margin so color never interferes with corner detection.

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10. Coordinate & Layout Strategy

1. **Page setup** — resolve page size → usable area = page minus margins minus quiet zone.
2. **Fixed blocks placed first**: header banner (logo + branding), student info grid, registration number — fixed heights regardless of exam config.
3. **Remaining area** split into N columns (default 2 for ≤60Q, 3 for >100Q).
4. **Sections flow column-major** in configured order; splits at question boundaries only, never mid-row.
5. **Per-section height** = header_bar_height (8mm, name-only) + (question_count × row_height).
6. **Reflow triggers**: question_count, marks_per_question, option_count, section order/name, page size, or student field list changes re-run the full layout pass.

Bubble coordinates: `x = col_start_x + option_index × bubble_spacing`, `y = section_start_y + header_height + question_index × row_spacing` — same formula the PDF generator and the scanner both use.

11. Machine-Readable JSON Schema

```
{
  "template_id": "string",
  "version": "integer",
  "exam_name": "string",
  "brand_colors": { "primary": "#123E7C", "accent": "#E8720C" },
  "page": {
    "size": "A4 | A5 | Letter",
    "orientation": "portrait | landscape",
    "margins_mm": { "top": 0, "bottom": 0, "left": 0, "right": 0 },
    "quiet_zone_mm": 0
  },
  "registration_marks": [
    { "id": "tl|tr|bl|br", "x_mm": 0, "y_mm": 0, "size_mm": 4, "shape": "square_filled"
  ],
  "header": {
    "school_name": "RK English Medium School",
    "exam_name": "string", "year": 0,
    "logo_ref": "string|null", "subtitle": "string|null",
    "extra_line": "string|null"
  },
  "student_fields": [
    { "field_id": "string", "label": "string", "order": 0, "input_type": "text|numeric
      "required": true, "enabled": true, "x_mm": 0, "y_mm": 0, "width_mm": 0 }
  ],
  "registration_number": { "digits": 6, "box_size_mm": {"w":8,"h":10}, "gap_mm": 1, "or
  "paper_set": { "enabled": true, "options": ["A","B","C","D"], "bubble_diameter_mm": 4
  "qr": { "enabled": false, "x_mm": 0, "y_mm": 0, "size_mm": 16, "encodes": ["template_
  "layout": { "columns": 2, "column_gap_mm": 10 },
  "sections": [
    {
      "section_id": "string", "name": "string", "order": 0,
      "allocated_marks": 0, "question_count": 0, "marks_per_question": 0,
      "option_count": 4, "question_numbering": { "start": 0, "end": 0 },
      "negative_marking": { "rule": "none | per_question | every_n_wrong", "n": 0, "dec
      "unanswered_marks": 0,
      "bubble": { "diameter_mm": 4, "spacing_mm": 8, "row_spacing_mm": 8 }
    }
  ],
  "scanner_params": { "dpi": 300, "color_mode": "grayscale", "device_profile": "RICOH_I
  "recognition_thresholds": { "fill_ratio_marked": 0.65, "fill_ratio_ambiguous_band": 0
}
```

12. Template Versioning

- Version bumps only on scanner-critical geometry changes — registration marks, bubble coordinates, QR position, page size.
- Color, logo, school name, header text — cosmetic, no version bump needed.
- Old versions never overwritten; printed sheets stay scannable.

13. Scanner Mapping & RICOH IM 2700 Ingestion

> RICOH IM 2700 → batch scan (200–300 DPI, grayscale) → detect template via QR → orientation + perspective correction (4 registration marks) → warp to canonical A4 space → load template config by template_id + version → sample bubble fill at the same x/y formula used for generation → fill-ratio threshold → answer payload → marking engine

The scanner reads bubble fill in grayscale regardless of the brand colors on the printed page — color only exists in the banner/section-bar regions the recognition engine never samples.

14. Marking Engine

```
{"correct": 1, "wrong_group_size": 3, "wrong_group_penalty": 1, "unanswered": 0}
```

Every 3 wrong = -1, applied by the evaluation engine, never printed on the sheet.

15. Dashboard Model & Validation Rules

Rule	Blocks PDF generation?
Σ(question marks in section) ≠ section allocated_marks	Yes — hard error
Σ(section allocated_marks) ≠ exam total_marks	Yes — hard error

question_count = 0 in an active section	Yes
option_count < 2	Yes
Registration/QR/bubble/logo regions overlap	Yes — hard error
Content exceeds printable margin or quiet zone	Yes — hard error
Section reordering	No — triggers reflow + review warning
Duplicate section_id or question_no overlap	Yes — hard error

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16. Print & Scanner Specification

- A4 portrait, vector PDF, 100% scale, no auto-fit.
- Brand colors (blue banner, orange accents) print in full color; everything the scanner samples (bubbles, registration marks, question numbers) stays pure black on white.
- RICOH IM 2700: 200–300 DPI, grayscale ADF — color banner converts cleanly to grayscale for the scan pass since it never overlaps a sampled region.

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17. Example RISE 2027 Config

```
{
  "template_id": "RISE-OMR-v1",
  "version": 1,
  "exam_name": "RISE 2027 - RK Integrated Scholarship Examination",
  "brand_colors": {"primary":"#123E7C","accent":"#E8720C"},
  "header": {"school_name":"RK English Medium School","exam_name":"RISE 2027","subtitle
  "page": { "size": "A4", "orientation": "portrait",
    "margins_mm": {"top":8,"bottom":8,"left":8,"right":8}, "quiet_zone_mm": 4 },
  "student_fields": [
    {"field_id":"name","label":"Student Name","order":1,"input_type":"handwritten","req
    {"field_id":"class","label":"Class","order":2,"input_type":"handwritten","required"
    {"field_id":"section","label":"Section","order":3,"input_type":"handwritten","requi
    {"field_id":"date","label":"Date","order":4,"input_type":"handwritten","required":t
    {"field_id":"phone","label":"Phone Number","order":5,"input_type":"handwritten","re
    {"field_id":"reg_no","label":"Registration Number","order":6,"input_type":"handwrit
  ],
  "paper_set": {"enabled":true,"options":["A","B","C","D"],"bubble_diameter_mm":4},
  "qr": {"enabled": false},
```

```
  "layout": { "columns": 2, "column_gap_mm": 10 },
  "sections": [
    {"section_id":"telugu","name":"TELUGU","order":1,"allocated_marks":3,"q
    {"section_id":"hindi","name":"HINDI","order":2,"allocated_marks":3,"que
    {"section_id":"english","name":"ENGLISH","order":3,"allocated_marks":5,
    {"section_id":"maths","name":"MATHEMATICS","order":4,"allocated_marks"
    {"section_id":"science","name":"SCIENCE","order":5,"allocated_marks":18,
    {"section_id":"social","name":"SOCIAL STUDIES","order":6,"allocated_mar
    {"section_id":"gk_computer","name":"GK / COMPUTER","order":7,"allocated
    {"section_id":"current_affairs","name":"CURRENT AFFAIRS","order":8,"all
  ],
  "marking_engine_rule": {"correct":1,"wrong_group_size":3,"wrong_group_pen
  "total_marks": 50,
  "total_questions": 50,
  "scanner_params": {"dpi":300,"color_mode":"grayscale","device_profile":"R
}
```


Still no reference image attached in this chat — colors, logo slot, and spacing above are built to your written spec. Upload the actual reference and I'll match it exactly.